The doctrinal Herbrand's theorem and its Stone dual

See recording at `https://youtu.be/VWTmOX9Tn84`

Marco Abbadini

Université catholique de Louvain, Belgium

6th ItaCa Workshop, Milan, Italy, 22 December 2025

Joint work in progress with Francesca Guffanti

1. Herbrand's theorem

Herbrand's theorem (1930) describes the validity of an existential formula $\exists x\, \phi(x)$ (with $\phi(x)$ <u>quantifier-free</u>) in terms of the validity of quantifier-free formulas:

$$\mathcal{T} \vdash \exists x\, \phi(x) \iff \text{there are } c_1, \dots, c_n \text{ s.t. } \mathcal{T} \vdash \phi(c_1) \vee \cdots \vee \phi(c_n).$$

It holds modulo any universal theory $\mathcal{T}$ (i.e.: made of universal closures $\forall \underline{x}\, \alpha(\underline{x})$ of q.f. formulas, such as the theory of preorders, the theory of partial orders, any quasivariety of algebras).

2. First-order Boolean doctrines

Classical **propositional** logic

$\Downarrow$

Boolean algebras

Classical **first-order** logic

$\Downarrow$

First-order Boolean doctrines

(Lawvere, 1969)

3. Stone duality

Classical **propositional** logic	Classical **first-order** logic
$\Downarrow$	$\Downarrow$
Boolean algebras	First-order Boolean doctrines
$\cong^{\mathrm{op}}$	$\cong^{\mathrm{op}}$
Stone spaces	Polyadic spaces (Joyal, 1971)

It is a duality between syntax (algebra of formulas) and semantics (spaces of models).

Aims

I will quickly address the question:

- What is the doctrinal reading of Herbrand's theorem? (algebras of formulas)

to then turn to the fun part:

- <u>What is its Stone dual?</u> (spaces of models)

The doctrinal reading of Herbrand's theorem

First-order Boolean doctrines

A *first-order Boolean doctrine (over* $\mathsf{FinSet}^{\mathrm{op}}$*)* is a functor

$$\mathbf{P} \colon \mathsf{FinSet} \to \mathsf{BA}$$

such that "adding dummy variables has a right and left adjoint" ($\forall$ and $\exists$), which moreover commute with substitutions (Beck-Chevalley).

For the talk: one sort, no function symbols, no $=$.

For a first-order theory $\mathcal{T}$:

$\mathbf{P}(X) = \{$FO formulas with free variables in $X\}/\mathcal{T}$-interprovability.

$\mathbf{P}$ on morphisms: simultaneous substitutions.

A bit more on Herbrand's theorem

Herbrand's theorem: for a universal theory $\mathcal{T}$:

$\mathcal{T} \vdash \exists x\, \phi(x) \iff$ there are $c_1, \ldots, c_n$ s.t. $\mathcal{T} \vdash \phi(c_1) \vee \cdots \vee \phi(c_n)$.

It <u>describes the validity</u> modulo a <u>universal theory</u> $\mathcal{T}$ of any formula of <u>quantifier alternation depth ≤ 1</u>, i.e.

$$\bigwedge \bigvee \alpha(\underline{z}),$$

with $\alpha(\underline{z})$ of the form

$$\exists \underline{x}\, \phi(\underline{x}, \underline{z}), \qquad \forall \underline{y}\, \psi(\underline{y}, \underline{z}),$$

with ϕ and ψ quantifier free.

Herbrand's theorem in doctrinal form

- Start with a **Boolean doctrine** (i.e. just a functor)

$$\mathbf{P} \colon \mathsf{FinSet} \to \mathsf{BA},$$

  modeling the class of **quantifier-free formulas** modulo a universal theory $\mathcal{T}$.

- $\mathbf{P}$ has a **quantifier completion**

$$\mathbf{P} \hookrightarrow \mathbf{P}^{\forall\exists},$$

  the universal way of freely adding first-order quantifiers. $\mathbf{P}^{\forall\exists}$ models the class of **all first-order formulas** modulo $\mathcal{T}$.

Herbrand's theorem in doctrinal form

- "Formulas of **quantifier alternation depth** ≤ 1":

$$\mathbf{P} \hookrightarrow \mathbf{P}_1 \hookrightarrow \mathbf{P}^{\forall\exists}$$

  $\mathbf{P}_1$ = freely adding to $\mathbf{P}$ a layer of quantifier alternation depth.

- **Herbrand's thm** = explicit description of $\mathbf{P}_1$ *in terms of* $\mathbf{P}$.

A., Guffanti. Freely adding one layer of quantifiers to a Boolean doctrine. On arxiv. (2024)

See also

Wrigley. Existential completions and Herbrand's theorem. On arxiv. (2025)

Doctrinal Herbrand's thm: first step in the description of "free" first-order Boolean doctrines via <u>iterative addition of nested quantifiers</u>.

My interest in the Stone dual of Herbrand's thm

1. **An open problem.**

   Pitts' 1992: is every Heyting algebra the poset of subterminals of some topos?

   Pataraia announced a positive answer using

   $$\underline{\textit{iterative addition of nested quantifiers}} + \underline{\textit{duality}},$$

   but passed away before having left enough details.

   We start to develop the technology in the setting of classical first-order logic.

My interest in the Stone dual of Herbrand's thm

2. **An invitation.**

   We've been inspired by Gehrke's talk at CT 20→21 (Genoa):

   doctrinal perspective + complexity of formulas + duality.

   (Gehrke, Jakl, Reggio.)

3. **A success case.**

   Dualities simplify free constructions: Ghilardi used

   iterative addition of nested implications + *duality*

   to describe free finitely generated Heyting algebras.

The Stone dual of Herbrand's theorem

Stone duality and doctrines

Composing a first-order Boolean doctrine

$$\mathbf{P} \colon \mathsf{FinSet} \to \mathsf{BA}$$

with Stone duality

$$\mathsf{BA} \cong \mathsf{Stone}^{\mathrm{op}}$$

gives...

A *polyadic space (over* $\mathsf{FinSet}^{\mathrm{op}}$*)* is a functor

$$\mathbf{E} \colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$$

with the following properties.

- Openness ($\leftrightarrow$ existence of adjoints).
- Amalgamation ($\leftrightarrow$ Beck-Chevalley).

Going back to

Joyal. Polyadic spaces and elementary theories. (1971)

See also Marquès' PhD thesis and

van Gool, Marquès. On duality and model theory for polyadic spaces. (2024)

Given a first-order theory $\mathcal{T}$ in a relational language, we have

$$\begin{aligned} \mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} &\longrightarrow \mathsf{Stone} \\ X &\longmapsto \mathrm{Mod}_X(\mathcal{T})_{/\equiv_{\mathrm{FO}}} \end{aligned}$$

with

$$\mathrm{Mod}_X(\mathcal{T}) := \{(M, \nu) \mid M \text{ model of } \mathcal{T},\ \nu\colon X \to M \text{ map}\}$$

where $(M, \nu) \equiv_{\mathrm{FO}} (M', \nu')$ if they are *elementarily equivalent*, i.e. they satisfy the same first-order formulas with free variables in X.

This is a polyadic space (and, vice versa, they are all of this form).

$$
\begin{tikzcd}
\mathrm{Mod}_{(-)}(\mathcal{T})_{/\equiv_{\mathrm{q.f.}}} & \mathrm{Mod}_{(-)}(\mathcal{T})_{/\equiv_{\mathrm{QA}\leq 1}} \arrow[l, two heads] & \mathrm{Mod}_{(-)}(\mathcal{T})_{/\equiv_{\mathrm{FO}}} \arrow[l, two heads] \\
\mathbf{E} \arrow[u, phantom, "\cong"] & \mathbf{E}_1 \arrow[l, two heads] \arrow[u, phantom, "\cong"] & \mathbf{E}^{\forall\exists} \arrow[l, two heads] \arrow[u, phantom, "\cong"] \\
\mathbf{P} \arrow[r, hook] \arrow[d, phantom, "\cong"] & \mathbf{P}_1 \arrow[r, hook] \arrow[d, phantom, "\cong"] & \mathbf{P}^{\forall\exists} \arrow[d, phantom, "\cong"] \\
\mathrm{Form}_{\mathrm{q.f.}}(-)_{/\dashv\vdash_{\mathcal{T}}} \arrow[r, hook] & \mathrm{Form}_{\mathrm{QA}\leq 1}(-)_{/\dashv\vdash_{\mathcal{T}}} \arrow[r, hook] & \mathrm{Form}_{\mathrm{FO}}(-)_{/\dashv\vdash_{\mathcal{T}}}
\end{tikzcd}
$$

- Let $\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ be a functor.
- Let $\mathbf{P}\colon \mathsf{FinSet} \to \mathsf{BA}$ be its Stone dual.

Herbrand's thm = description of $\mathbf{P}_1$ in terms of $\mathbf{P}$.

Stone dual of Herbrand's thm = descript. of $\mathbf{E}_1$ in terms of $\mathbf{E}$.

$$\mathbf{E}_1(\varnothing) \cong \mathrm{Mod}(\mathcal{T})_{/\equiv_{\mathrm{QA}\leq 1}}.$$

Theorem (The Stone dual of Herbrand's theorem)

$\mathbf{E}_1(\varnothing)$ *is the Stone space of* <u>*Herbrand types for* $\mathbf{E}$</u>.

Definition

A *Herbrand type for a functor* $\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ is a subfunctor of $\mathbf{E}$ mapping finite products (of $\mathsf{FinSet}^{\mathrm{op}}$) to quasi-products.

Quasi-product $:=$ the morphism to the product is epi.

Herbrand type: tuple $(F_X)_{X \in \mathsf{FinSet}}$, with $F_X \subseteq \mathbf{E}(X)$ closed, s.t.

1. $(F_X)_X$ is closed under substitution;
2. $F_{X_1 \sqcup X_2}$ is a quasi-product of F_{X_1} and F_{X_2};
3. $F_\varnothing$ is quasi-terminal (i.e., nonempty). $F_\varnothing$ is a singleton.

Theorem

Let $\mathcal{T}$ be a universal theory, and $\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ *the functor mapping X to*

$$\mathbf{E}(X) = \mathrm{Mod}_X(\mathcal{T})_{/\equiv_{\mathrm{q.f.}}} := \{(M, X \to M) \mid M \in \mathrm{Mod}(\mathcal{T})\}_{/\equiv_{\mathrm{q.f.}}}.$$

Then,

$$\mathrm{Mod}(\mathcal{T})_{/\equiv_{\mathrm{QA}\leq 1}} \cong \{\textit{Herbrand types for } \mathbf{E}\}.$$

$$[M]_{\equiv_{\mathrm{QA}\leq 1}} \longmapsto \left(X \mapsto \overline{\left\{ \left[(M, X \to M)\right]_{\equiv_{\mathrm{q.f.}}} \right\}} \right).$$

We have been inspired by a similar idea in

Gehrke, Jakl, Reggio. A cook's tour of duality in logic: from quantifiers, through Vietoris, to measures. (2025)

To sum up

The Stone dual of Herbrand's theorem: Given a functor

$$\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone},$$

we describe

$$\mathbf{E}_1\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone},$$

(freely adding to $\mathbf{E}$ one layer of QA). In particular:

$$\mathbf{E}_1(\varnothing) = \{\text{Herbrand types for } \mathbf{E}\}$$

Herbrand type: subfunctor, finite products $\mapsto$ quasi-products.

- The Stone space $\mathbf{E}_1(X)$, for any X, is defined similarly.
- Not just $\mathsf{FinSet}^{\mathrm{op}}$, but any category with finite products (i.e., allowing function symbols and multiple sorts).
- With equality: same construction.

Thank you!

Appendix

Definition

Given a category C with finite products, a *first-order Boolean doctrine over* C is a functor $\mathbf{P}\colon \mathsf{C}^{\mathrm{op}} \to \mathsf{BA}$ with the following properties.

1. (Universal) For all $X, Y \in \mathsf{C}$,

$$\mathbf{P}(\mathrm{pr}_X^{X\times Y})\colon \mathbf{P}(X) \to \mathbf{P}(X \times Y)$$

   has a right adjoint $(\forall Y)_X$.

2. (Beck-Chevalley) For any $f\colon X' \to X$,

$$\begin{tikzcd}
X & \mathbf{P}(X \times Y) \arrow[r, "(\forall Y)_X"] \arrow[d, "\mathbf{P}(f\times \mathrm{id}_Y)"'] & \mathbf{P}(X) \arrow[d, "\mathbf{P}(f)"] \\
X' \arrow[u, "f"] & \mathbf{P}(X' \times Y) \arrow[r, "(\forall Y)_{X'}"'] & \mathbf{P}(X').
\end{tikzcd}$$

A *polyadic space (over* $\mathsf{FinSet}^{\mathrm{op}}$*)* is a functor

$$\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$$

s.t.

- Openness ($\leftrightarrow$ adjoints): for all $X, Y \in \mathsf{FinSet}$, $\mathbf{E}(X \hookrightarrow X \sqcup Y)\colon \mathbf{E}(X \sqcup Y) \to \mathbf{E}(X)$ is an open map.
- Amalgamation ($\leftrightarrow$ Beck-Chevalley): For any $f\colon X \to X'$,

$$\begin{tikzcd}
\mathbf{E}(X' \sqcup Y) \arrow[r, "\mathbf{E}(X' \hookrightarrow X' \sqcup Y)"] \arrow[d, "\mathbf{E}(f \sqcup \mathrm{id}_Y)"'] & \mathbf{E}(X') \arrow[d, "\mathbf{E}(f)"] \\
\mathbf{E}(X \sqcup Y) \arrow[r, "\mathbf{E}(X \hookrightarrow X \sqcup Y)"'] & \mathbf{E}(X)
\end{tikzcd}$$

is a quasi pullback (= pullback up to epi).

Theorem

Let $\mathcal{T}$ be a universal theory, and $\mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ *the functor mapping X to*

$$\mathbf{E}(X) = \mathrm{Mod}_X(\mathcal{T})_{/\equiv_{\mathrm{q.f.}}} := \{(M, X \to M) \mid M \in \mathrm{Mod}(\mathcal{T})\}_{/\equiv_{\mathrm{q.f.}}}.$$

Then,

$$\mathrm{Mod}(\mathcal{T})_{/\equiv_{\mathrm{QA}\leq 1}} \cong \{\textit{Herbrand types for } \mathbf{E}\}.$$

$$[M]_{\equiv_{\mathrm{QA}\leq 1}} \longmapsto \left(X \mapsto \overline{\left\{ \left[(M, X \to M)\right]_{\equiv_{\mathrm{q.f.}}} \right\}} \right).$$

We found this idea in

Gehrke, Jakl, Reggio. A cook's tour of duality in logic: from quantifiers, through Vietoris, to measures. (2025)

1. $(F_X)_X$ is closed under substitution: For every $f\colon Y \to X$,

$$\begin{tikzcd} F_X \arrow[r, dashed] & F_Y \\ \mathbf{E}(X) \arrow[u, phantom, "\rotatebox{90}{$\subseteq$}"] \arrow[r, "\mathbf{E}(f)"] & \mathbf{E}(Y) \arrow[u, phantom, "\rotatebox{90}{$\subseteq$}"] \end{tikzcd}$$

$\mathbf{E}(f)[F_X] \subseteq F_Y$.

Idea: An element in F_X is more or less of the form $(M, X \xrightarrow{\nu} M)$.

$$\mathbf{E}(f)(M, X \xrightarrow{\nu} M) = (M, Y \xrightarrow{\nu \circ f} M) \in F_Y.$$

2. $F_{X_1 \sqcup X_2}$ is the quasi-product of F_{X_1} and F_{X_2}:

$$\begin{tikzcd}
 & & y \arrow[dll, dashed, mapsto] \arrow[drr, dashed, mapsto] & & \\
x_1 & & F_{X_1 \sqcup X_2} \arrow[ll] \arrow[rr] \arrow[u, phantom, "\in"] \arrow[d, phantom, "\cap"] & & x_2 \\
 & F_{X_1} \arrow[ul, phantom, "\in"] \arrow[dl] & \mathbf{E}(X_1 \sqcup X_2) \arrow[dl] \arrow[dr] & F_{X_2} \arrow[ur, phantom, "\in"] \arrow[dr] & \\
 & \mathbf{E}(X_1) \arrow[u, phantom, "\cap"] & & \mathbf{E}(X_2) \arrow[u, phantom, "\cap"] &
\end{tikzcd}$$

For all X_1, X_2, $x_1 \in F_{X_1}$ and $x_2 \in F_{X_2}$ there is $y \in F_{X_1 \sqcup X_2}$ s.t. $\mathbf{E}(X_1 \hookrightarrow X_1 \sqcup X_2)(y) = x_1$ and $\mathbf{E}(X_2 \hookrightarrow X_1 \sqcup X_2)(y) = x_2$.

Idea: x_1 is more or less of the form $[(M, X_1 \xrightarrow{\nu_1} M)]$, x_2 is more or less of the form $[(M, X_2 \xrightarrow{\nu_2} M)]$. Then, one can take

$$y = \left[\left(M, \begin{pmatrix} X_1 \xrightarrow{\nu_1} M \\ X_2 \xrightarrow{\nu_2} M \end{pmatrix} : X_1 \sqcup X_2 \to M\right)\right].$$

3. $F_\varnothing$ is a singleton.

Idea: it is $(M, \varnothing \to M)$.

The construction works also with equality in the language.

Empty language, with $=$, $\mathcal{T} := \{\forall x \forall y\, (x = y)\}$:

$$\begin{aligned} \mathbf{E} \colon \mathsf{FinSet}^{\mathrm{op}} &\longrightarrow \mathsf{Stone} \\ X &\longmapsto \mathrm{Mod}_X(\mathcal{T})_{/\equiv_{\mathrm{q.f.}}} \cong \{*\} \end{aligned}$$

(because every q.f. formula is equivalent to $\top$ or $\bot$).

Two classes of models wrt $\equiv_{\mathrm{QA}\leq 1}$:

1. the class of singleton models (satisfying $\exists x \top$).
2. the class of the empty model (satisfying $\neg \exists x \top$).

In fact, $\mathbf{E}$ has two Herbrand types:

1. $\mathbf{E}$;
2. $$\begin{aligned} \mathsf{FinSet}^{\mathrm{op}} &\longrightarrow \mathsf{Stone} \\ X &\longmapsto \begin{cases} \{*\} & \text{if } X = \varnothing; \\ \varnothing & \text{if } X \neq \varnothing. \end{cases} \end{aligned}$$

For the empty theory $\mathcal{T}$ in the empty language without "$=$",

$$\begin{aligned} \mathbf{E}\colon \mathsf{FinSet}^{\mathrm{op}} &\longrightarrow \mathsf{Stone} \\ X &\longmapsto \mathrm{Mod}_X(\mathcal{T})_{/\equiv_{\mathrm{q.f.}}} \cong \{*\} \end{aligned}$$

(as the only quantifier-free formulas are $\top$ and $\bot$).

Two equivalence classes of models wrt $\equiv_{\mathrm{QA}\leq 1}$:

1. the class of nonempty models (satisfying $\exists x \top$).
2. the class of the empty model (satisfying $\neg\exists x \top$).

In fact, $\mathbf{E}$ has two Herbrand types:

1. $\mathbf{E}$;
2. $$\begin{aligned} \mathsf{FinSet}^{\mathrm{op}} &\longrightarrow \mathsf{Stone} \\ X &\longmapsto \begin{cases} \{*\} & \text{if } X = \varnothing; \\ \varnothing & \text{if } X \neq \varnothing. \end{cases} \end{aligned}$$